Modeling Protein Sequence Adaptation to Environmental Factors Using Gold Database Samples and UniProt Sequences

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Abstract

This document describes the data acquisition, enrichment, feature engineering, modeling and evaluation plan for predicting protein sequence variation as a function of environmental factors. Samples (with locality information) from the Gold database will be linked to UniProt protein sequences. Locality coordinates will be enriched with environmental covariates (climate, soil, oceanographic, and anthropogenic metrics). The enriched dataset will be used to train conditional sequence prediction models that learn how environmental changes affect protein sequences. This writeup is designed to be kept up-to-date as the project progresses and to serve as the authoritative project PDF.

1 Background and Motivation

Many extremophiles and environment-adapted organisms exhibit sequence features correlated with environmental pressures (temperature, salinity, pH, radiation, etc.). By systematically linking curated sample metadata (location, substrate, temperature recorded at sampling) from the Gold dataset to protein sequences in UniProt, and enriching each sample with high-resolution environmental covariates derived from the sampling coordinates, we can build models that predict sequence-level changes in response to environmental factor perturbations.

2 Objectives

1. Gather Gold database samples with precise geocoordinates and linked specimen identifiers.
2. Retrieve corresponding protein sequences (or sequence identifiers) from UniProt for each sample where available.
3. Enrich each sample with environmental covariates derived from the geographic coordinates.
4. Construct feature pipelines to combine environment data and sample metadata, and construct labels from sequence representations.
5. Train conditional sequence prediction models to predict sequence variation under controlled changes to environmental inputs.
6. Evaluate model fidelity with sequence-based metrics and ecological plausibility checks.

3 Data Sources and Retrieval

3.1 Gold database (samples and metadata)

- Primary source for sample metadata: sample identifier, sampling date, latitude/longitude, sampled substrate, measurement metadata (if available).

- Retrieval: use Gold API or downloadable TSV/CSV export. Filter for records with valid coordinates and sample-level identifiers that can be linked to sequence records.

3.2 UniProt (sequences)

- UniProt entries are retrieved using UniProt accession numbers/TAX ID or by mapping sample identifiers (where the Gold entry contains links to sequence accessions or organism identifiers).
- Retrieval methods: UniProt REST API, batch queries, or local UniProt downloads. Use accession mapping endpoints or TAX ID to translate between gene or sample IDs and protein sequences when needed.

3.3 Linking strategy

- Exact match: if Gold provides UniProt accession or EMBL/GenBank accessions, use them directly to fetch canonical protein sequences.
- Taxonomic + gene matching: when direct accessions are missing, use taxon and locus/gene information to identify candidate UniProt entries; record confidence and provenance for each mapping.

4 Environmental Enrichment (Work in Progress)

Given a latitude/longitude for each sample, enrich with the following covariates. Prefer open datasets with programmatic access and clear spatial resolution metadata.

4.1 Static and long-term climatologies

- WorldClim / CHELSA: long-term monthly climatologies (temperature, precipitation, bioclimatic variables).
- SoilGrids: soil pH, organic carbon, texture classes.

4.2 Short-term / time-specific weather and radiation

- NASA POWER / ERA5 (or ERA5-Land): surface temperature, radiation, humidity for the sampling date (if available) or for a representative window.

4.3 Oceanographic variables (for marine samples)

- Marine datasets: sea-surface temperature (SST), salinity, pH proxies, chlorophyll-a (from MODIS/Aqua or OISST), and depth/bathymetry.

4.4 Anthropogenic and land-use covariates

- Human footprint indices, land cover classes, proximity to built-up areas and agricultural extent (from MODIS/ESA Land Cover products).

4.5 Implementation notes for enrichment

- For raster datasets (WorldClim, SoilGrids, etc.) use a spatial sampling library (rasterio / xarray / rioxarray in Python) to extract point values at sample coordinates. Handle coordinate reference systems explicitly.

- For model-ready time-series (e.g., NASA POWER), align the sampling date to the environmental record and compute summary statistics (mean, min, max, anomalies) over relevant windows (sampling day, 7-day, 30-day, seasonal).
- Record data provenance and resolution (e.g., WorldClim v2 2.5 arc-min).

5 Feature Engineering (Work in Progress)

5.1 Environmental feature vector

- Numeric variables: temperature metrics, precipitation, salinity, pH, radiation, soil organic matter, elevation, distance-to-coast, etc.
- Categorical variables: land cover class, substrate type (rock, sediment, biofilm), biome.
- Normalization: scale continuous variables (z-score or quantile scaling) and encode categorical variables with one-hot or learned embeddings.

5.2 Sequence label representation

We consider multiple label formulations depending on modeling goals:

1. Residue-level prediction: predict the amino-acid at each aligned position. Requires multiple-sequence alignment (MSA) and fixed-length positional labels (useful for single-protein families).
2. K-mer / signature prediction: predict distribution over k-mer counts or sequence-derived summary statistics (GC-equivalent, hydrophobicity indexes).
3. Embedding/regression targets: map sequences to continuous embeddings (from pre-trained protein language models) and predict embedding vectors as a regression conditioned on environmental features.
4. Generative sequence model: learn conditional sequence generator (transformer, conditional VAE) that outputs full amino-acid sequences given environmental features.

5.3 Choice trade-offs

- Alignment-based residue prediction is interpretable but requires clean MSAs and homologous sequences.
- Embedding/regression approaches are flexible for diverse sequence lengths and can leverage pre-trained protein language models (e.g., ESM, ProtBert).

6 Modeling Approaches (Work in Progress)

6.1 Conditional discriminative models

- Predictors for residue or embedding targets: feed-forward networks, gradient-boosted trees (for aggregated labels), or sequence-to-label CNN/LSTM models when inputs include local sequence context.

6.2 Conditional generative models

Sequence generation conditioned on environment using

- Conditional Transformer decoder (conditioning via concatenated tokens or cross-attention to environment embeddings).

- Conditional VAE: latent space captures sequence variation and decoder reconstructs sequences given environment vectors.

6.3 Model inputs and conditioning

- Simple conditioning: concatenate normalized environment vector to each token embedding or to the decoder's initial states.
- Cross-attention: allow decoder layers to attend to a learned environment embedding sequence.

7 Training Pipeline and Reproducibility

7.1 Data pipeline

Steps:
1 Ingest Gold sample rows with coordinates;
2 Map to UniProt sequences;
3 Enrich coordinates with environmental rasters/APIs;
4 Precompute sequence labels and embeddings;
5 Construct training/validation splits stratified by geography, taxon, and sampling time where appropriate.

7.2 Implementation notes

- Organize code under 'src/' with modular components: 'data/' (downloaders, mappers), 'features/' (environment extraction, normalization), 'models/', and 'experiments/' (training harness, hyperparameter configs).
- Save provenance metadata (source, raster name, extraction timestamp) in a companion JSON or Parquet table for each sample.

7.3 Reproducibility

- Use pinned package versions in 'requirements.txt' or 'environment.yml'.
- Provide a 'README.md' with commands to reproduce the dataset enrichment and training. Store random seeds and checkpoint model weights in 'checkpoints/'.

8 Evaluation and Validation

8.1 Quantitative metrics

- Per-residue accuracy, per-sequence edit distance (Levenshtein), BLEU-style scores adapted for amino-acids, and embedding-space cosine similarity.
- Calibration metrics for probabilistic models (Brier score, negative log-likelihood).

8.2 Ecological and biological validation

- Compare predicted sequences' biophysical properties (e.g., predicted thermostability, isoelectric point, hydrophobicity) against expected trends for environmental changes.
- Case studies: hold out geographically or environmentally distinct regions and test whether predicted sequence adjustments are biologically plausible.

9 Experiments and Expected Results

- Baseline: predict sequence embeddings with a linear model from environmental covariates; measure embedding reconstruction error.
- Generative baseline: conditional LSTM/Transformer that generates short protein regions given environment vectors.
- Ablations: remove subsets of environmental factors (temperature, salinity) to quantify their marginal effects on sequence predictions.
- Sensitivity analysis: perturb input covariates and measure sequence changes to estimate model responsiveness.

10 Data Privacy, Ethics, and Limitations

- Track and respect any access restrictions or licensing for Gold or UniProt subsets. If geolocation data are sensitive, consider obfuscation or aggregation to larger spatial units.
- Limitations: observational confounding (correlated environmental variables, unobserved sampling biases), phylogenetic dependence among samples, and the challenge of causally attributing sequence changes to environment.

11 Next Steps

- Implement automated mapping between Gold and UniProt and store a provenance table.
- Prototype environmental extraction for a small sample set (10–100 points) to validate raster/API choices.
- Build a baseline embedding regression and evaluate.

12 Appendix: Recommended packages and tools

- Python: ‘requests‘, ‘pandas‘, ‘geopandas‘, ‘rasterio‘, ‘xarray‘, ‘rioxarray‘, ‘scikit-learn‘, ‘pytorch‘/‘tensorflow‘, ‘biopython‘.
- Data sources: UniProt REST API, WorldClim, CHELSA, SoilGrids, NASA POWER, ERA5, MODIS/Ocean products.

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